

Fagus grandifolia growth and mortality a decade after the emergence of Beech leaf disease

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ABSTRACT

Beech leaf disease (BLD) is poised to cause major declines in American beech (*Fagus grandifolia*) across the eastern United States and parts of Canada. Given the dominance of this tree, quantifying impacts of this emerging disease is critical. Using long-term data from an existing field experiment (originally established to explore the impacts of soil chemistry on forests), we quantify rates of mortality and growth in American beech afflicted with BLD near the disease's epicenter. Since the initial observation of BLD in Cuyahoga County, OH in 2014, 75 of the 263 (29 %) American beech trees within our study have died. Most of this mortality was recent, with 2021–2023 displaying the highest levels of beech mortality (56 trees dying across the three years). Mortality was distributed unevenly across size classes, with the highest rates of mortality occurring in trees <25 cm DBH. Mortality rates were influenced by beech abundance, suggesting that areas with high concentrations of beech may experience higher rates of mortality. In addition, beech grew more slowly in recent years (2017–2022), suggesting a probable slowing of growth rates associated with BLD. Further, we observed lower growth rates in plots with the addition of soil amendments, but only before the arrival of BLD. As a dominant tree in many forests, this decline in American beech could catalyze larger stand-level changes in forest composition and function as BLD persists on the landscape and continues to spread into new areas.

1. Introduction

American beech (*Fagus grandifolia*) is a common (and often dominant) deciduous tree species in eastern North American forests (Tubbs and Houston, 1990). With a range extending from Nova Scotia in Canada to northern Florida in the United States and west into Texas, the American beech is a constituent of 20 forest cover types and is a dominant tree in Sugar Maple-Beech-Yellow Birch and Beech-Sugar Maple forests (Tubbs and Houston, 1990). Easily identifiable by its smooth light gray bark, American beech is a climax species growing slowly within forest understories prior to ascending into the canopy (Runkle, 1981).

The future of American beech is uncertain due to a new threat to the species. Beech leaf disease (BLD) is an emerging forest disease affecting trees in the genus *Fagus*. First identified in Lake County, OH, USA in 2012, BLD has since spread to 15 states across the eastern United States and into the Ontario province of Canada (Cleveland Metroparks, 2023). Symptoms have been linked to a non-native nematode *Litylenchus crenatae* ssp. *mccannii*, which infests beech buds prior to leaf out (Carta et al.,

2020; Reed et al., 2020). Early symptoms include dark green interveinal banding easily seen from the underside of the leaf. More severe symptoms include fully darkened leaves that may be curled and/or shrunk (Ewing et al., 2019). Changes in leaves are both anatomical and physiological (McIntire, 2023; Vieira et al., 2023). Thickening of leaves is the result of increased thickness of the spongy mesophyll, palisade layer, and abaxial epidermis (Fletcher et al., 2024). Stomatal density is approximately 50 % lower in symptomatic leaves, possibly due to disruptions in cell division during leaf development as a result of BLD (Fletcher et al., 2024). Maximum photosynthetic rate of symptomatic leaves can be approximately 60 % lower than asymptomatic leaves, potentially leading to a reduction in carbon assimilation capacity (Fletcher et al., 2024). Such changes could potentially lead to mortality as a result of a reduction in stored carbon.

The potential for BLD to reduce the abundance of American beech within North American forests is particularly problematic as the species' wide distribution and frequent dominance have made it both ecologically and economically important across large swaths of its range. American beech is a nut-producing species, making it a critical food

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source for a variety of wildlife from mice to black bears (Coladonato, 1991). It is common for sections of the trunk to become hollow, providing an ideal den for many animals such as squirrels and raccoons. Wood from American beech trees is utilized as a building material, used most often in flooring, furniture, and turned products (Coladonato, 1991). It is also a top choice for fuelwood (Tubbs and Houston, 1990).

In addition to pests and diseases, including BLD, forests are also shaped by extrinsic factors that drive key forest parameters, such as soil chemistry. Acid rain deposition, for example, has a long history of altering forest form and function through long-term shifts in soil pH and nutrient availability (Burke et al., 2021). Such changes have the potential to influence species interactions (Brown et al., 2001; Finzi et al., 1998; van Breemen et al., 1997) including interactions between hosts and diseases (Cale et al., 2015). Acid deposition, and the accompanying loss of base cations such as Ca, has the potential to alter plant physiology and reduce plant growth. Ca specifically has a role in defense and repair functions within plants, and reduction or reduced supply of Ca could alter plant pest and disease resistance (McLaughlin and Percy, 1999). Reduced uptake and acquisition of essential nutrients, whether Ca, P or K, has been linked to plant susceptibility to pests and diseases (Bala et al., 2018). It is uncertain how BLD incidence responds to underlying differences across the landscape in soil chemistry, including changes in chemistry due to acid deposition. As novel diseases like BLD emerge, it is important to understand various contextualities that may influence their impacts.

Mortality of American beech has been reported in areas with high levels of BLD infestation (Ewing et al., 2019; Martin and Volk, 2022; Reed et al., 2022), though rates of mortality associated with the disease have yet to be formally quantified. Here, we quantify rates of mortality within the epicenter of BLD using annually collected data predating the presence of BLD. We also explore growth rates before and after BLD arrival. Data used in this study were originally part of a long-term experimental study of phosphorus availability and soil acidity on the function and composition of microbial communities (DeForest et al., 2012). As such, we also explore the effects of soil chemistry on growth and mortality of American beech in association with BLD.

2. Materials & methods

2.1. Site description and experimental design

We explored long-term shifts in growth and mortality of American beech in the context of an already-existing experiment into the impacts of soil acidification on forest communities. Utilizing this existing manipulative experiment allowed us access to data on tree growth and mortality before the emergence of BLD. Research took place at Case Western Reserve University's Squire Valleeview and Valley Ridge Farm (Squire Valleeview Farm) in Hunting Valley, Cuyahoga County, OH, USA (41.49° N, -81.42° W) as part of a research project aimed at exploring the impacts of phosphorus availability and soil acidification on temperate forests (DeForest et al., 2012). The site, located on the Allegheny plateau, is characterized by a naturally occurring mixed hardwood forest dominated by American beech (*Fagus grandifolia*), white oak (*Quercus alba*), red oak (*Quercus rubra*), and sugar maple (*Acer saccharum*). The regional climate is continental, characterized by hot, humid summers and cold, snowy winters. The average annual temperature for 2010–2023 was 11.4 °C with an average annual precipitation of 113.3 cm ("NOWData," 2024). BLD was first recorded in Cuyahoga County, OH in 2014, two years after its initial discovery in neighboring Lake County, OH. BLD symptoms were readily observed on trees at Squire Valleeview Farm by 2017 (Koonce, personal communication). By 2023, all American beech included in the study area exhibited symptoms of BLD (personal observation). Notably, beech bark disease has not been observed within the plots (personal observation).

In 2009, twelve research plots were established at Squire Valleeview Farm. In a randomized complete block design, plots were treated with

one of four soil chemistry manipulations: Control (no soil additions), Low Acidity (limestone addition), High Phosphorous (triple super phosphate addition), and Low Acidity + High Phosphorus (limestone and triple super phosphate addition). Research plots were 20 m x 40 m (800 m²). These four treatments were spatially arranged into three blocks. Phosphorous and pH levels were regularly checked within the plots, and soils were amended as necessary to maintain treatments (see Deforest et al., 2012 and Burke et al., 2021 for further detail).

2.2. Field data collection

Diameter at breast height (DBH) was measured for all living American beech trees 5 cm DBH and above within the research plots – a total of 263 American beech trees – from 2010 through 2023. Annual measurements were taken in early summer. Trees were noted as dead or alive at the time of the annual survey. To characterize mortality across size classes, each tree was assigned to one of five size classes based on DBH using the Overstory Tree Diameter Class table from the U.S. Forest Service (Forest Service, 2023): sapling (2.54 cm < 12.45 cm), pole (12.7 cm < 25.15 cm), small (25.4 cm < 50.55 cm), medium (50.8 cm < 75.95 cm), or large (> 76.2 cm). Due to the existing experimental design, this study did not analyze mortality of trees <5 cm DBH, which includes some of the saplings in the plots, as well as the seedlings.

2.3. Analyses

We displayed annual mortality rates as both the number of beech trees that died in each year, as well as cumulative beech mortality. Cumulative mortality was modeled exponentially. We also displayed annual mortality rates by size class. Due to the uneven distribution of trees across size classes, mortality rates for individual size classes were displayed as the percentage of newly dead individuals in the size class out of the total number of beech trees in that size class in that year. Size class was newly determined for each tree in each year of the study to account for tree growth. Note that since DBH was not recorded for dead trees, DBH for the last recorded living year was used to assign size class at time of death. Mortality across soil treatments was also calculated and displayed as the percentage of newly dead individuals in treatment out of total number of beech trees in that treatment.

To predict probability of mortality as a factor of tree size, year, plot-level beech abundance, and soil treatment, we ran a binomial generalized linear model (GLM). We represented beech abundance as the total basal area of American beech per plot. This was calculated by determining the basal area for each individual tree ($\text{dbh}^2 \times 0.00007854$ [Bettinger et al., 2017]), and then summing the basal area for all individuals within each plot. We fit the model using the glm function as follows: (Mortality ~ DBH + year + soil treatment + beech basal area). Response was calculated for trees on a 0.5 increment size for all years and all treatments.

To understand shifts in growth rates associated with BLD, we calculated the annual growth of individual beech trees as the percent increase in DBH using: $((\text{final DBH} - \text{initial DBH}) / \text{initial DBH}) \times 100$. We limited our analysis of growth to trees that were alive for the entire study duration of 2011–2023. We compared average growth over two 6-year time periods: 2011–2016 and 2017–2022, representing the time periods before and after the first observations of widespread symptoms of BLD at the site. We then calculated an average growth rate for each individual for each of the two 6-year time periods. We used a Wilcoxon signed rank test to compare differences between growth rates across the two time periods since data were not normally distributed. To understand if soil chemistry modified disease impacts on tree growth, we ran a mixed effects model with growth as the response variable, time period and soil treatment as interacting fixed effects, and individual tree as a random effect. Data were not normally distributed, and so were log transformed. Type III sums of squares were used to account for unbalanced sample sizes across treatments. All analyses were run in RStudio

(ver. 2023.06.1) (R Core Team, 2023).

3. Results

3.1. Mortality and size classes

Mortality of American beech in Squire Valleeview Farm increased over the course of the study, with the largest increases occurring in the three most recent years (Fig. 1). Between 2011 and 2023, 78 of the 263 (29.6 %) beech trees (including saplings) died within the study plots. The vast majority (75 trees or ~96 %) of this mortality occurred since the initial documentation of BLD in the county in 2014. From 2011 through 2015, annual mortality of beech stood at <1 %. Cumulative mortality increased following an exponential trend ($\text{Adj. } R^2 = 0.9671$; $\ln(\text{Mortality}) = -650.75 + 0.32(\text{Year})$) over 13 years (Fig. A.1 & Table A.1). A rise in mortality was documented beginning in 2016, approximately two years after BLD arrived in the county, starting relatively low at a rate of about 1 % mortality (Table A.3). 2021–2023 displayed the highest levels of beech mortality with 56 trees dying across three years, consisting of 71.8 % of total beech deaths across the entire study period. 2022 exhibited the highest mortality, with 27 trees dying in that year (34 % of total dead, 10 % of total beech).

Approximately half of the beech trees within the study plots were classified as saplings (133–179 trees in this size class, depending on year). The second most abundant size class was poles (36–43 trees), followed by small trees (24–27 trees), medium trees (8–12 trees), and finally large trees (2–3 trees) (Fig. A.2). The number of trees within a size class shifted from year-to-year as a result of both tree death and growth (into the next size class). Saplings exhibited mortality annually since 2016, but showed a steady increase in recent years, with the highest mortality seen in 2022, at 14 % (Fig. 2). Mortality in the pole size class followed a similar pattern, with mortality first observed in 2017 and occurring in every subsequent year with the exception of 2020. The mortality among poles was highest in 2023, at 8 % (Fig. 2). Mortality among small trees began in 2021, 7 years after the introduction of BLD, with mortality observed in all subsequent years. Medium trees were the only group to exhibit mortality prior to the introduction of BLD in the county, with mortality observed in 2011 (2 trees) and 2013 (1 tree), as well as after BLD introduction in both 2019, and 2021. Only 3 trees within the research plots were classified as large, and no mortality was documented among these trees.

The number of dead trees varied among soil treatments. Control plots had the highest mortality (29 trees, or 37 %) while the High Phosphorus (TSP) plots had the lowest mortality (12 trees, or 18 %) (Fig. A.3 & Table A.3).

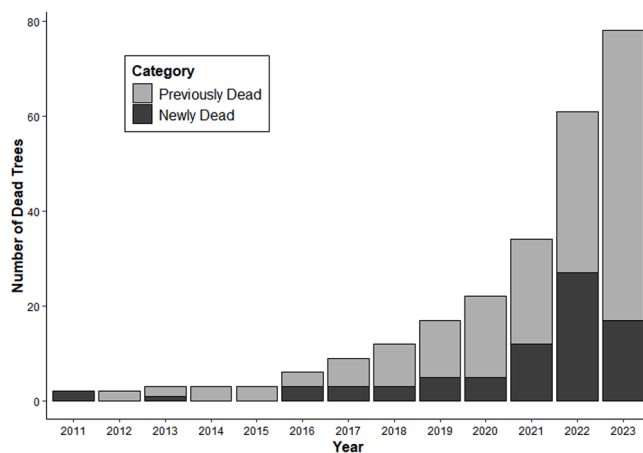


Fig. 1. Cumulative number of dead American beech. Black segments of the bars indicate the number of beech newly dead in that year while the gray segments indicate the trees that died in previous years (beginning in 2011).

Tree size (as measured by DBH) was a significant predictor of tree mortality ($Z = -2.923$; $p = 0.003$) across all soil treatments and years, with smaller trees exhibiting a higher probability of death than larger trees (Fig. 3). Year was a statistically significant predictor of mortality ($Z = 8.164$, $p < 0.01$), with mortality increasing over time. Beech abundance (represented by plot-level beech basal area) was a significant predictor of beech mortality ($Z = 4.506$, $p < 0.01$) with the probability of mortality increasing with increasing beech abundance. Soil amendments proved to have a marginally significant impact on mortality, with lime-treated plots having a marginally lower probability of mortality than unamended plots ($Z = -1.916$, $p = 0.05$) (Fig. A.4 & Table A.4).

3.3. Growth

Beech growth, as measured by percent change in DBH, was lower in 2017–2022 (0.46 %) relative to 2011–2016 (1.77 %) ($W = 724,835$, $p\text{-value} < 0.01$) (Fig. 4). Time period and soil treatment both had a significant impact on growth ($\chi^2 = 61.923$, $Df = 1$, $p\text{-value} < 0.01$; $\chi^2 = 31.857$, $Df = 3$, $p\text{-value} < 0.01$) (Fig. 5). The interaction between time period and soil treatment was significant in driving growth ($\chi^2 = 24.372$, $Df = 3$, $p\text{-value} < 0.01$) (Fig. 5). From 2011–2016 (before widespread BLD symptoms), plots with any addition of limestone (Lime and Lime + TSP) exhibited lower growth (1.26 % and 1.44 %) relative to control (2.09 %) and TSP (2.15 %) plots ($\chi^2 = 31.857$, $Df = 3$, $p\text{-value} < 0.01$). Once symptoms of BLD were widespread in the site (2017–2022), the soil treatment showed little to no effect on growth (Table A.5).

4. Discussion

Since the arrival of beech leaf disease in Cuyahoga County, OH in 2014, approximately 30 % of American beech in our long-term research plots at Squire Valleeview Farm have died. Notably, mortality was most extensive among saplings, relative to larger size classes. This trend supports other reports of elevated mortality of saplings in the presence of BLD within the region (Volk et al., 2019). While three beech trees died within the study plots prior to the arrival of BLD in the county of study in 2014, no mortality was observed from that point until 2016. Notably, mortality was observed every year beginning in 2016, with a minimum of three trees dying each year from that point on. 2021 (seven years after the arrival of BLD within the county) saw the start of larger increases in mortality (12, 27, and 17 trees dying in 2021, 2022, and 2023, respectively). Healthy American beech tend to be long lived, reaching upwards of 300 years (Tubbs and Houston, 1990). Annual mortality rates of uninfected beech may be around 1.5 % (McWilliams et al., 2005), though was considerably lower within our study site before the arrival of BLD. Our findings suggest that BLD drives mortality in beech and are consistent with observations of the onset of BLD-induced mortality within 2–7 years of initial infestation (Martin and Volk, 2022). Additionally, results indicate the potential for an exponential increase in mortality within the first decade of infestation.

Since the first observation of BLD in 2012, the disease has spread to 15 states in the continental United States and into Ontario, Canada (Cleveland Metroparks, 2023). This rapid expansion and widespread distribution could lead to significant impacts on deciduous forests, particularly if the disease drives high rates of mortality. As BLD is still a relatively new forest pest (<13 years since initial observation), long-term patterns of mortality are still emerging. Because of the proximity of the study site to the epicenter of BLD, these forest stands help us understand expected mortality rates caused by BLD over the first decade of infestation and beyond. We observed the highest rates of mortality in smaller trees. There were annual deaths in trees in the sapling and pole size classes beginning in 2016, just two years after the first documentation of BLD in Cuyahoga County. This finding aligns with others that have shown that the onset of mortality is quick, and mortality rates are high in saplings (Ewing et al., 2019; Martin and Volk, 2022; Volk et al., 2019). Saplings, in general, are more susceptible to changes in

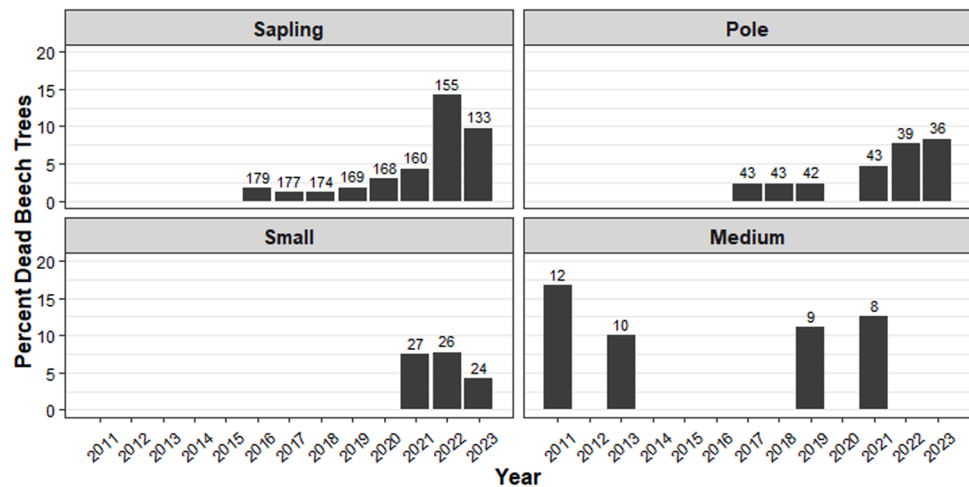


Fig. 2. Percent of beech trees newly dead per size class in each year of monitoring. The “Large” size class is not included as there were relatively few individuals that fell in this class and none of these trees died over the course of this study. Numbers on bars indicate the total number of trees within the size class in a given year.

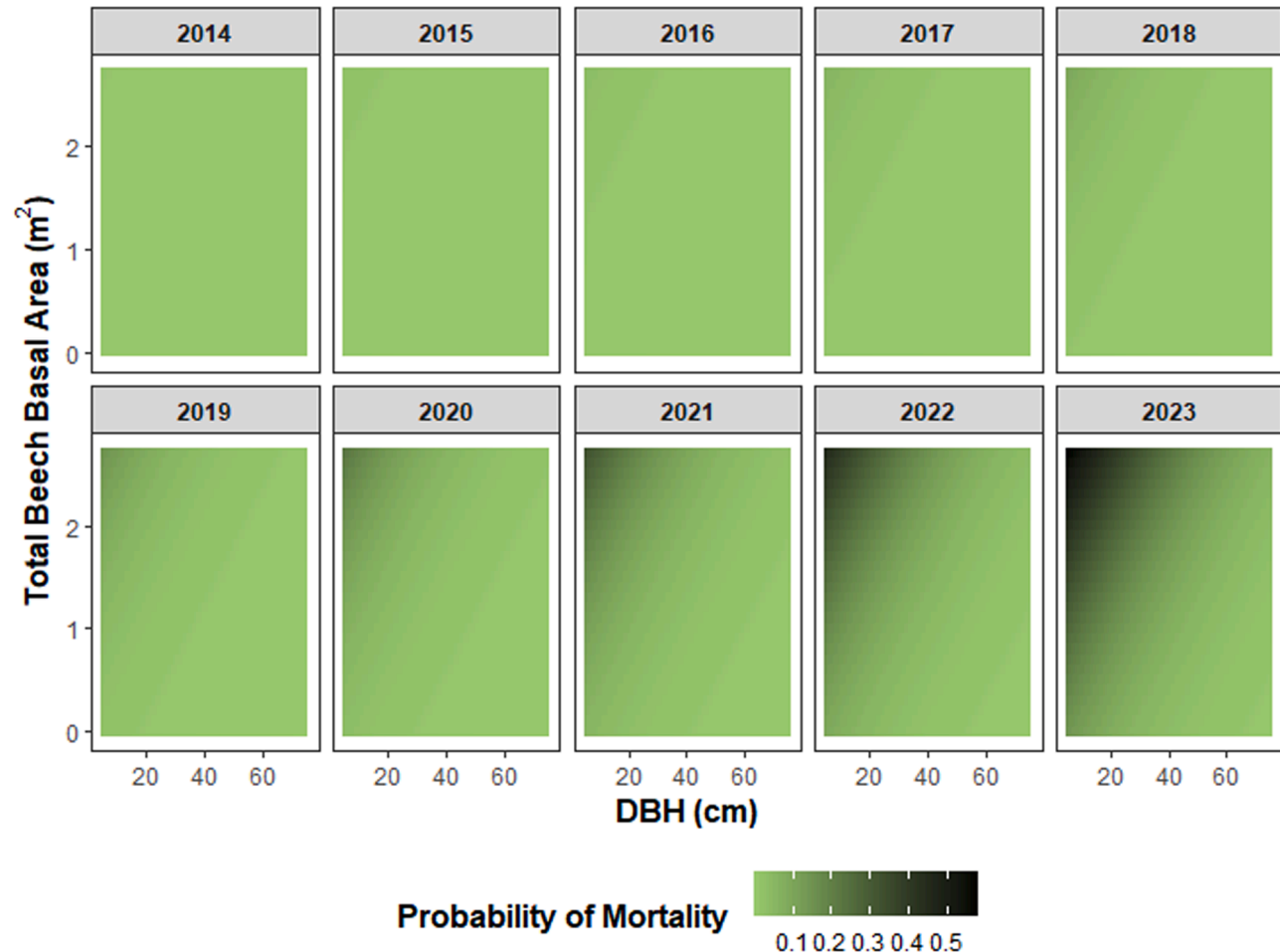


Fig. 3. Heat map displaying the predicted probability of mortality for American beech as a function of tree size, year, and plot-level beech basal area. Light green signifies low probability of mortality while black signifies higher probability of mortality. Year was included as a continuous variable, encompassing the year of arrival of BLD in the county (2014) through the final year of data collection (2023). The minimum and maximum recorded DBH of trees and beech basal area within plots served as the upper and lower limits.

precipitation and soil moisture than larger trees (Hanson et al., 2001). Predisposed vulnerability paired with impacts from BLD could explain high mortality among small trees. Since we did not analyze mortality of beech seedlings, future work should further investigate mortality in this

size class to continue understanding potential for BLD to alter forest composition.

We found the predicted mortality to be higher in areas with higher beech abundance (as measured by basal area). In a study of ash mortality

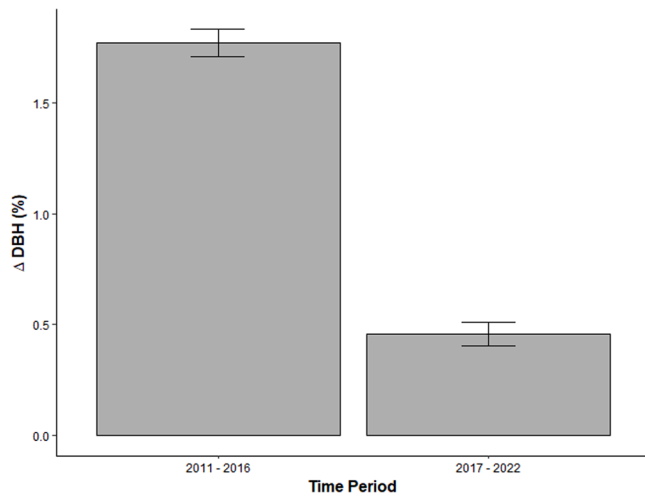


Fig. 4. Mean beech growth as measured by percent increase in DBH (%) from 2011 to 2016 and 2017–2022 $\pm$ standard error.

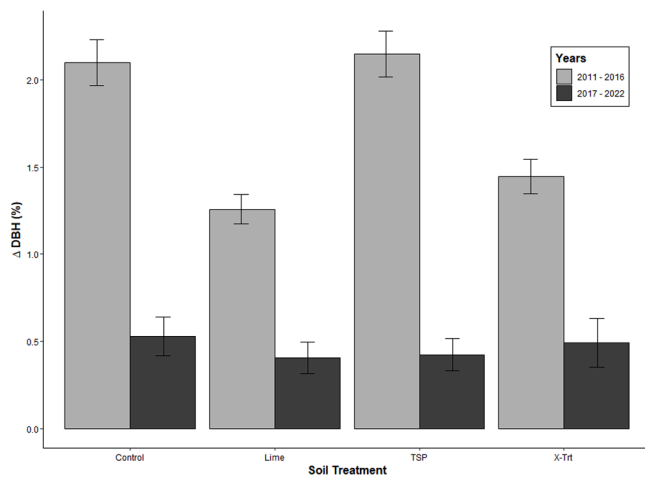


Fig. 5. Tree growth, as measured by change in DBH (%) for 2011–2016 and 2017–2022, grouped by soil treatment and averaged by plot $\pm$ standard error.

driven by emerald ash borer (EAB) (*Agrilus planipennis*), Smith et al. (2015) found no relationship between mortality and ash basal area in stands severely impacted by EAB. One explanation for our observed correlation between beech mortality and basal area could be increased intraspecific competition within forest stands, potentially leading to enhanced vulnerability. Studies of general tree mortality have documented that when levels of intraspecific competition are high as a result of elevated tree densities within a species, a thinning phase can occur. Mortality is often directly related to initial stem densities during this phase (Peet and Christensen, 1987). Fletcher et al. (2024) identified decreased photosynthetic rates paired with increased thickness of leaf tissues of BLD-symptomatic leaves. Such stress associated with BLD could interact with high levels of competition to drive mortality, particularly in smaller trees. Another mechanism driving the correlation between stand-level beech densities and mortality could be that higher beech densities facilitate transmission of the causal nematode among individuals. An association between high host abundances and increased rates of pest spread was identified in the case of the pest insect hemlock woolly adelgid and its host *Tsuga species* (Morin et al., 2009). The association between beech density and mortality may have important management implications. The exact mechanism(s) for the spread of *L. crenatae* are, of yet, unknown. That said, any future management recommendations will benefit from additional survey work, likely over

larger spatial scales, combined with experimental work to understand causative interactions between the disease and environment.

Rates of mortality among American beech trees within these BLD-impacted research plots appear somewhat slower compared to mortality rates imposed by several other important forest diseases and pests within North American forests. For example, emerald ash borer (EAB) can result in 85 % mortality in ash trees within 3–5 years of initial infestation (Kashian and Witter, 2011). Similarly, hemlock woolly adelgid (HWA) can result in mortality within 5–6 years, reaching 90 % mortality within 10 to 12 years in some cases (Mayer et al., 2002). Oak wilt disease has proven variable in its impacts across species within the genus *Quercus*. Northern red oaks (*Quercus rubra*) have been shown to be severely impacted, causing death within a year of infection (Haigh et al., 2011). American beech are also impacted by beech bark disease (BBD), caused by a non-native scale insect and a fungal pathogen (Witter et al., 2005). This disease impacts larger trees (>20 cm DBH) and has been documented to kill up to 50 % of trees within 10 years in heavily impacted areas (Kibbe and Bonello, 2018; Miller-Weeks, 1983). Concerningly, both small and large trees are vulnerable to disease when BBD and BLD are both present in a stand, with large trees disproportionately impacted by BBD and small trees disproportionately impacted by BLD. Given BBD's propensity to generate dense thickets of small beech trees (Cale et al., 2013) and BLD's significant impacts on beech saplings occurring at high densities, how these two diseases will interact remains unclear. Major compositional shifts, specifically the advancement of uneven-aged conditions, are suspected to occur in beech stands impacted by both BLD and BBD (Reed et al., 2022). These shifts could prove challenging for land managers focusing on conservation, resilience, and ecosystem services.

In addition to driving mortality of beech trees, BLD also seems to slow growth rates of living trees. We found growth of beech trees to be higher in the six years prior to onset of widespread BLD symptoms (2011–2016) than in the six years after widespread BLD symptoms were observed (2017–2022). Tree growth is commonly depressed by forest pests and pathogens (Kozłowski, 1969). For example, trees infected with beech bark disease show reductions in tree growth of 20 – 40 % compared to uninfected trees, and these impacts scale with disease severity (Gavin and Peart, 1993; Mize and Lea, 1979). Eastern hemlock (*Tsuga Canadensis*) severely infested with HWA can express a spike in growth immediately following infestation, followed by marked decrease growth and then mortality (Davis et al., 2007). Given the anatomical and physiological changes associated with BLD, the observed slowing of growth rates is a logical consequence.

Reductions in photosynthetic rates, higher construction costs of leaves, and early senescence could all be contributing to both the reduction in growth in later years of BLD progression and increased mortality. As a foliar pest, *L. crenatae* causes damage to leaves prior to leaf out, entering and feeding on the bud (Carta et al., 2020; Vieira et al., 2023). As a result, BLD-impacted leaves express decreases in photosynthetic rates (Fletcher et al., 2024) coupled with anatomical changes in leaves, indicating potential reallocation of energy to leaf tissues. In addition, severely symptomatic leaves can be dropped from trees as early as June (Fearer et al., 2022). Diseased beech may be redirecting (increasingly limited) resources to prioritize leaf development over growth. Further, diseased trees also often exhibit reductions in canopy cover, further reducing whole-tree photosynthetic capacity, likely contributing to decreased growth and ultimately mortality. Reductions in mycorrhizal colonization of highly symptomatic beech trees (Bashian-Victoroff et al., 2023) may additionally hinder nutrient uptake (Finlay, 2008). In advanced BLD, this combination of factors likely contributes to the observed mortality. While other notable forest pests and pathogens impact the xylem and phloem of plants disrupting the natural transport of nutrients, water, or other organic molecules needed for photosynthesis (Flower et al., 2013; Havill et al., 2016; Kibbe and Bonello, 2018), BLD has not been documented to alter the transport of water or nutrients. As such, the comparatively slower onset of mortality

and potentially lower rates of mortality associated with BLD are likely explained, at least to some extent, by the specific mechanisms by which *L. crenatae* impacts the functioning of beech trees.

While the exploration of soil chemistry was not the original intention of this study, soil chemistry proved a potentially interesting modulating factor. Specifically, after taking into account variable levels of beech abundance across the plots (as measured by basal area), limestone addition was marginally predictive of beech mortality, with limestone-treated plots expressing a lower mortality rate than controls. An important caveat here is that beech abundance also varied across soil treatments with limestone and TSP-treated plots having the lowest cumulative beech basal areas. Given the importance of beech abundance as a driver of disease mortality, this may influence our findings. Future work investigating the mechanisms underlying the relationships between soil amendment and mortality is warranted. While we observed differences in beech growth rates across soil treatments prior to widespread occurrence of BLD (with reduced growth rates observed in the lime-treated plots), there were no impacts of soil treatments on beech growth after the arrival of BLD. The addition of limestone may result in conditions less favorable for beech, generally slowing growth rates (van Breemen et al., 1997). Reduced response of growth to soil treatments later in the experiment could demonstrate inability of beech trees to respond effectively to soil chemistry in the presence of BLD, with growth rates already substantially reduced due to disease.

5. Conclusion

Long-term monitoring near the epicenter of beech leaf disease reveals marked increases in American beech mortality from before emergence of the disease to nine years after arrival of BLD in the study area. Although BLD cannot definitively be linked as the cause of mortality of beech at the site, results of this study suggest a strong association between disease arrival and the onset of increasing rates of mortality. We observed an exponential increase in beech death following BLD introduction, with saplings proving most vulnerable. Further, growth rates significantly declined within the study plots after the onset of disease. This research provides additional evidence that BLD poses a substantial threat to North American forests (Ewing et al., 2019; Martin and Volk, 2022; Reed et al., 2022). Continued monitoring within the epicenter of BLD will provide additional data on potential shifts in mortality and growth as we move into the second decade of BLD infestation. The state of beech within North American forests should be carefully monitored across the disease range as BLD continues to spread into new regions.

CRedit authorship contribution statement

Brianna L. Shepherd: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **David J. Burke:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Conceptualization. **Katharine L. Stuble:** Writing – review & editing, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.tfp.2025.100836.

Data availability

Data will be made available on request.

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